Embedding and Vector Database

1. Embedding models and selection rationale

Three back-ends share one interface (fit, encode, save). The default 'lsa' embedder is TF-IDF (1-2 grams, sublinear tf, English stop words) followed by TruncatedSVD [4]; it needs no download. 'tfidf' keeps the full sparse vocabulary as a dense vector; 'sentence-transformers' wraps a pretrained model (default all-MiniLM-L6-v2, documented output size 384 dimensions [2]). Rationale: the build environment could not reach HuggingFace, so a pretrained model could not be downloaded; LSA is the strongest embedding that could be fitted locally and executed. The sentence-transformers back-end is implemented but was NOT executed or tested, and no result in this document comes from it.

2. Dimensions, similarity metric and normalisation

Verified from the persisted index (manifest.json): embedder 'lsa', dimension 44, 45 vectors. The dimension is min(128, n_chunks - 1, features - 1); with 45 chunks it is capped at 44. The 'tfidf' embedder has 1795 dimensions. Every vector (stored and query) is L2-normalised, so the FAISS inner product equals cosine similarity; a test checks that stored vectors have unit norm.

3. Vector store, index construction and persistence

faiss.IndexFlatIP performs exact brute-force search [3]; at 45 vectors this is instantaneous and avoids approximation error. artifacts/index/ contains index.faiss, chunks.jsonl (text and metadata, in vector order), manifest.json (embedder, dimension, metric, chunk size/overlap, SHA-256 of every source file, counts, build time), embedder.joblib (fitted TF-IDF/SVD) and ingest_report.json. Loading verifies that index and chunk table have the same length and reports corruption as a user-safe error.

4. Query embedding, top-k retrieval and filtering

The query is stripped, embedded with the persisted embedder (same model as the index), and searched for the top-k (default 4) neighbours. Results are (rank, cosine score, chunk). An optional document filter scores all vectors and keeps the top-k matching chunks (exact on a flat index). Empty/None queries, k < 1 and query/index dimension mismatch raise RetrievalError with a clear message; any other failure is wrapped without a stack trace.

5. Duplicates and index updates

Chunks with identical lower-cased text are de-duplicated before indexing (first kept). Updating the knowledge base: add or edit files and press 'Rebuild index' (or run scripts/build_index.py). The manifest hashes make a stale index detectable, and the UI warns. Updates are a full rebuild because the LSA embedder must be refitted on the new corpus; a pretrained embedder would allow incremental adds but that is not implemented.

6. Configuration

RAG_EMBEDDER (lsa | tfidf | sentence-transformers), RAG_LSA_DIMS, RAG_ST_MODEL, RAG_CHUNK_SIZE, RAG_CHUNK_OVERLAP, RAG_TOP_K, RAG_MIN_SCORE (see .env.example).

7. Experiment: chunk size, top-k and embedder

Setup (scripts/run_experiments.py): the 26 evidence-labelled evaluation cases; overlap fixed at 20% of chunk size; each configuration builds its own index; Precision@k, Recall@k, MRR and Hit@k as defined in the Testing document. Actual output:

embedder	chunk	overlap	#chunks	k	P@k	R@k	MRR	Hit@k	passed
lsa	250	50	82	1	0.808	0.750	0.808	0.808	29/34
lsa	250	50	82	3	0.333	0.904	0.885	0.962	29/34
lsa	250	50	82	5	0.200	0.904	0.885	0.962	29/34
lsa	500	100	45	1	0.808	0.750	0.808	0.808	28/34
lsa	500	100	45	3	0.346	0.923	0.885	0.962	29/34
lsa	500	100	45	5	0.223	0.942	0.885	0.962	29/34
lsa	800	160	40	1	0.808	0.750	0.808	0.808	28/34
lsa	800	160	40	3	0.372	0.962	0.885	0.962	29/34
lsa	800	160	40	5	0.238	1.000	0.894	1.000	29/34
tfidf	250	50	82	1	0.808	0.750	0.808	0.808	29/34
tfidf	250	50	82	3	0.321	0.865	0.865	0.923	29/34
tfidf	250	50	82	5	0.192	0.865	0.865	0.923	29/34
tfidf	500	100	45	1	0.808	0.750	0.808	0.808	28/34
tfidf	500	100	45	3	0.346	0.923	0.878	0.962	29/34
tfidf	500	100	45	5	0.223	0.942	0.878	0.962	29/34
tfidf	800	160	40	1	0.808	0.750	0.808	0.808	28/34
tfidf	800	160	40	3	0.359	0.942	0.878	0.962	29/34
tfidf	800	160	40	5	0.238	1.000	0.888	1.000	29/34

Findings. (a) Raising k from 1 to 3 lifts Hit@k from 0.808 to 0.923-0.962 for every configuration, and to 1.000 only for chunk size 800 at k = 5; precision falls as k grows because most questions have a single relevant chunk. (b) Chunk size changes results by at most one or two

questions of 26; 800-character chunks reach full recall at k = 5, but larger chunks also contain more of the labelled phrases, so the comparison is confounded and no size is shown to be superior. (c) LSA and TF-IDF are nearly indistinguishable here (LSA ahead by one question at chunk size 250, k = 3). (d) 'Cases passed' is 28-29/34 in all configurations, i.e. it is limited by the generator's abstention rule, not by retrieval. The sample is far too small for statistical claims; the trends match what is expected (recall up, precision down with k).